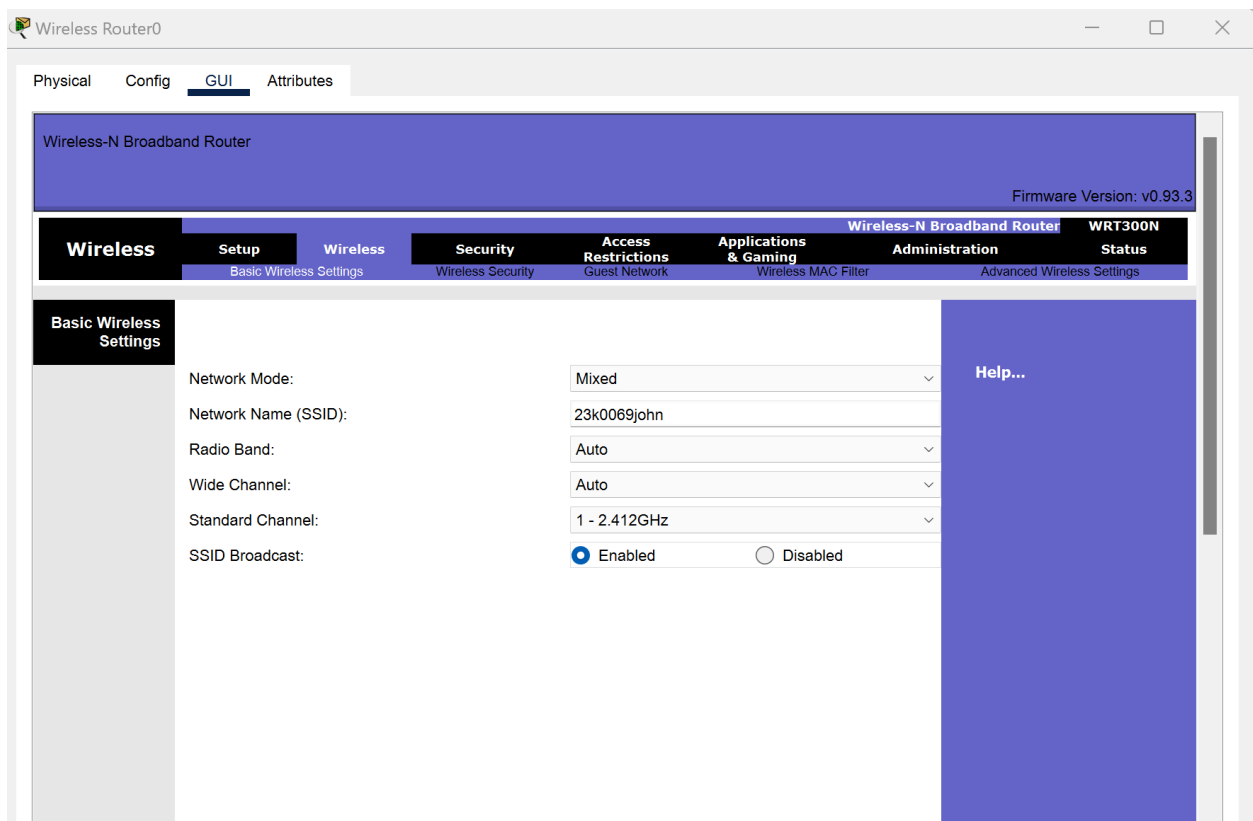


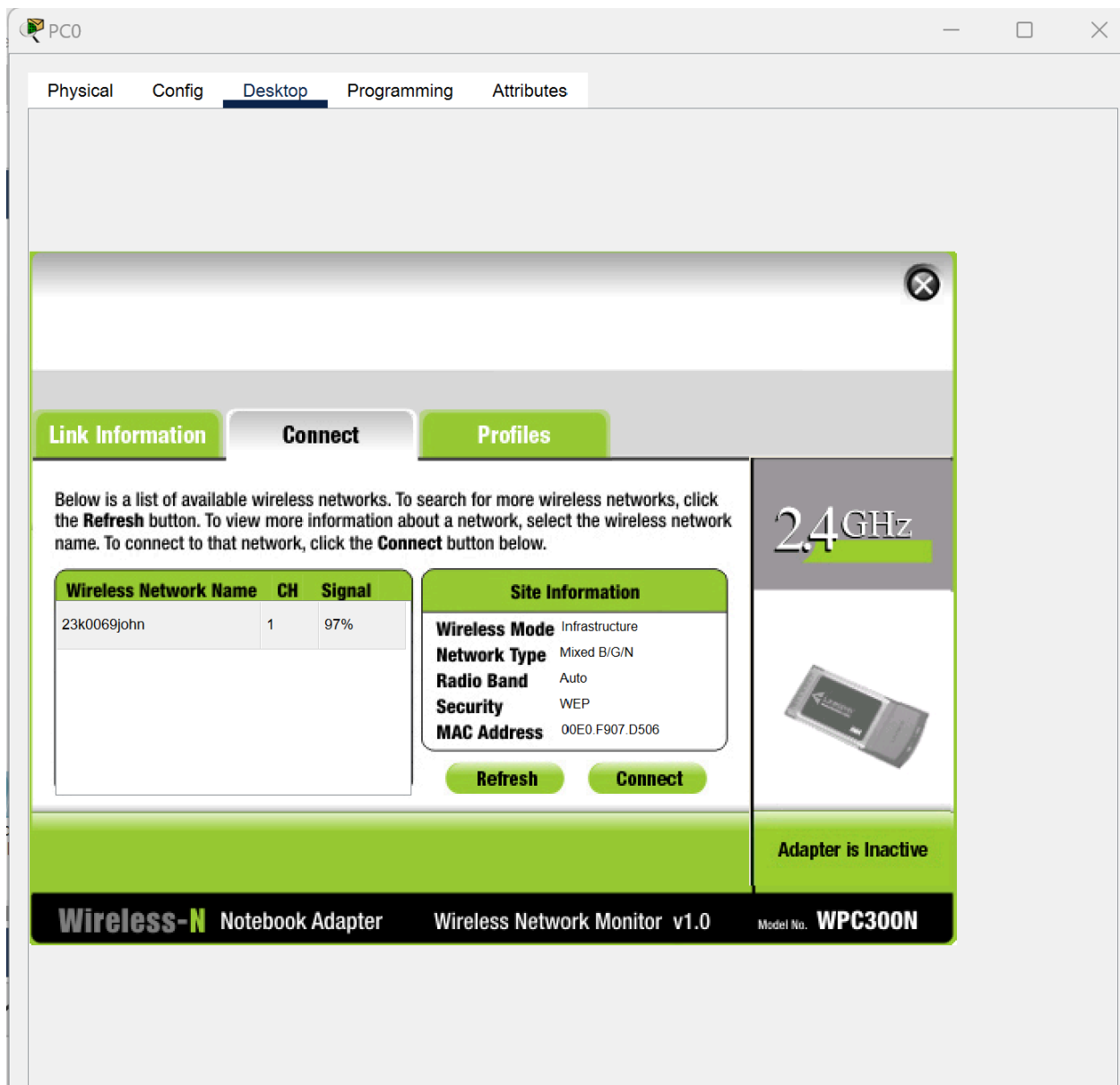
lab 9

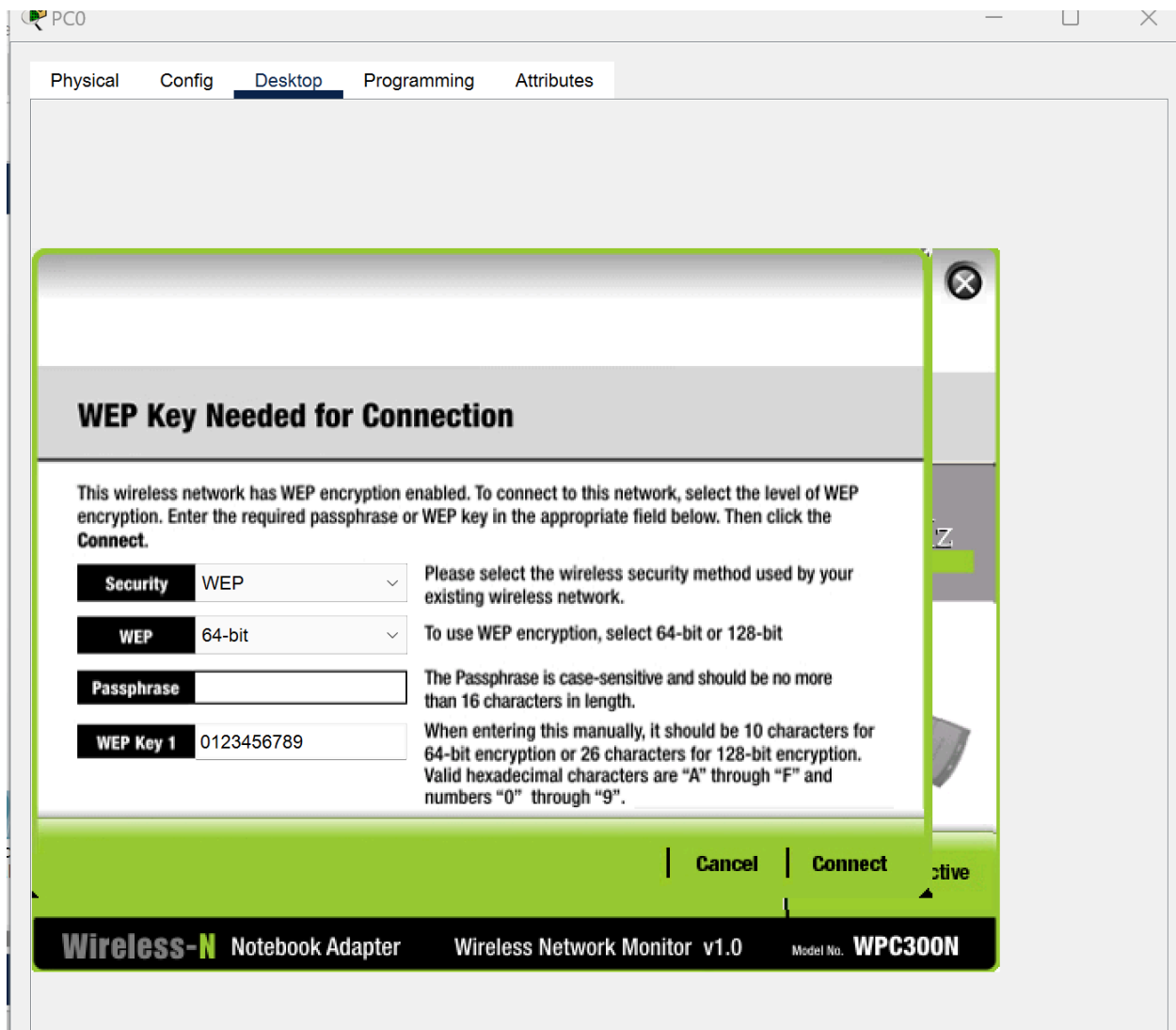
task 1

1.



2.Key: 0123456789





PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface Wireless0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.0.102

Subnet Mask 255.255.255.0

Default Gateway 192.168.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static Ipv6 request failed.

IPv6 Address /

Link Local Address FE80::201:97FF:FE44:766

Default Gateway

DNS Server

Link Information

Connect

Profiles

Below is a list of available wireless networks. To search for more wireless networks, click the **Refresh** button. To view more information about a network, select the wireless network name. To connect to that network, click the **Connect** button below.

Wireless Network Name	CH	Signal
23k0069john	1	87%

Site Information

Wireless Mode

Infrastructure

Network Type

Mixed B/G/N

Radio Band

Auto

Security

WEP

MAC Address

00E0.F907.D506

Refresh

Connect

2.4GHz



Adapter is Inactive

Wireless-N Notebook Adapter

Wireless Network Monitor v1.0

Model No. WPC300N

PC1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

Interface

Wireless0

IP Configuration

☒ DHCP

☐ Static

DHCP request successful.

IPv4 Address

192.168.0.103

Subnet Mask

255.255.255.0

Default Gateway

192.168.0.1

DNS Server

0.0.0.0

Tablet PC0

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

3G/4G Cell1

Bluetooth

Wireless0

Port Status

On

Bandwidth

300 Mbps

MAC Address

00D0.5891.8C0E

SSID

23k0069john

Authentication

☐ Disabled

☒ WEP

☐ WPA-PSK

☐ WPA

☐ 802.1X

☐ WPA2-PSK

☐ WPA2

Method:

WEP Key

0123456789

PSK Pass Phrase

User ID

Password

Method:

MD5

User Name

Password

Encryption Type

40/64-Bits (10 Hex digits)

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

169.254.140.17

Subnet Mask

255.255.0.0

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

/

Link Local Address:

FE80::2D0:58FF:FE91:8C0E

lab 9

6

Smartphone0

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

3G/4G Cell1

Bluetooth

Wireless0

Port Status

On

Bandwidth

300 Mbps

MAC Address

0000.0CD6.54D1

SSID

23k0069

Authentication

☐ Disabled

☒ WEP

☐ WPA-PSK

☐ WPA

☐ 802.1X

☐ WPA2-PSK

☐ WPA2

Method:

WEP Key

0123456789

PSK Pass Phrase

User ID

Password

Method

MD5

User Name

Password

Encryption Type

40/64-Bits (10 Hex digits)

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

169.254.84.209

Subnet Mask

255.255.0.0

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

Link Local Address: FE80::200:CFF:FED6:54D1

lab 9

7

Smartphone0

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

3G/4G Cell1

Bluetooth

Wireless0

Port Status

On

Bandwidth

300 Mbps

MAC Address

0000.0CD6.54D1

SSID

23k0069john

Authentication

☐ Disabled

☒ WEP

☐ WPA-PSK

☐ WPA

☐ 802.1X

☐ WPA2-PSK

☐ WPA2

Method:

WEP Key

0123456789

PSK Pass Phrase

User ID

Password

Method:

MD5

User Name

Password

Encryption Type

40/64-Bits (10 Hex digits)

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

169.254.84.210

Subnet Mask

255.255.0.0

IPv6 Configuration

☒ Automatic

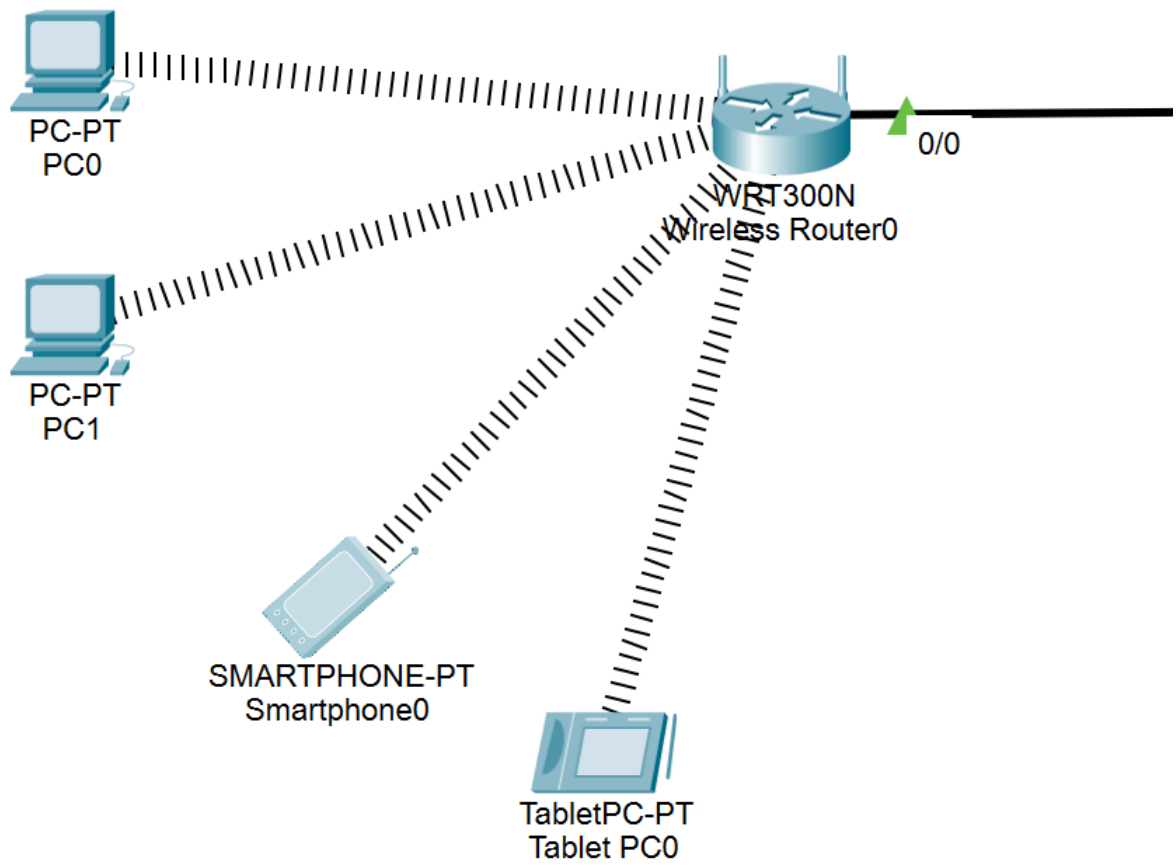
☐ Static

IPv6 Address

/

Link Local Address:

FE80::200:CFF:FED6:54D1



Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/0

Port Status

☒

On

Bandwidth

☒

1000 Mbps

☐

100 Mbps

☐

10 Mbps

☒

Auto

Duplex

☒

Half Duplex

☐

Full Duplex

☒

Auto

MAC Address0010.11A8.5B01

IP Configuration

IPv4 Address192.168.1.1

Subnet Mask255.255.255.0

Tx Ring Limit10

Equivalent IOS Commands

Router>enable

Server0

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.1.3

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

192.168.1.3

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

 /

Link Local Address

FE80::201:64FF:FE6C:7C94

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 192.168.1.3

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:FFFF:FE09:9775

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

3.SSH on router and switch

```
Router(config-if)#  
Router(config-if)#exit  
Router(config)#hostname r1  
r1(config)#enable pass pass  
r1(config)#ip domain-name a.b  
r1(config)#crypto key gen rsa  
The name for the keys will be: r1.a.b  
Choose the size of the key modulus in the range of 360 to 4096 for your  
General Purpose Keys. Choosing a key modulus greater than 512 may take  
a few minutes.  
  
How many bits in the modulus [512]: 24  
% A decimal number between 360 and 4096  
How many bits in the modulus [512]: 24  
% A decimal number between 360 and 4096  
How many bits in the modulus [512]: 28  
% A decimal number between 360 and 4096  
How many bits in the modulus [512]: 1024  
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]  
  
r1(config)#username admin secret pass  
*Mar 1 0:45:33.806: %SSH-5-ENABLED: SSH 1.99 has been enabled  
r1(config)#line vty 0 4  
r1(config-line)#login local  
r1(config-line)#trans in ssh  
r1(config-line)#exit  
r1(config)#ip ssh ver 2  
r1(config)#do wr  
Building configuration...  
[OK]  
r1(config)#
```

Copy

Paste

ENG

Switch0

PhysicalConfigCLIAttributes

IOS Command Line Interface

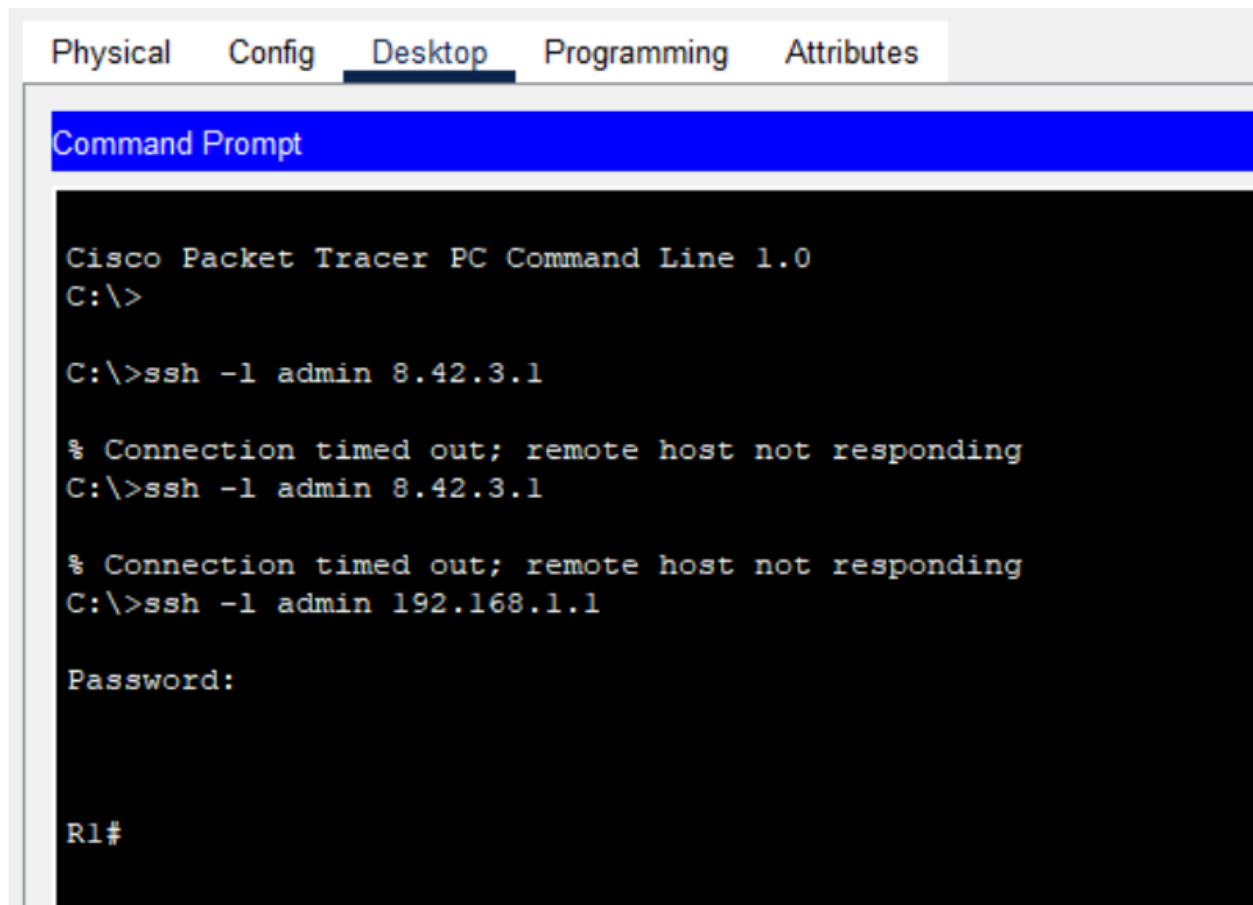
```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int vlan 1
Switch(config-if)#ip addr 192.168.1.9 255.255.255.0
Switch(config-if)#exit
Switch(config)#hos a
a(config)#enable pass pass
a(config)#ip domain-name a.b
a(config)#crypto key gen rsa
The name for the keys will be: a.a.b
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

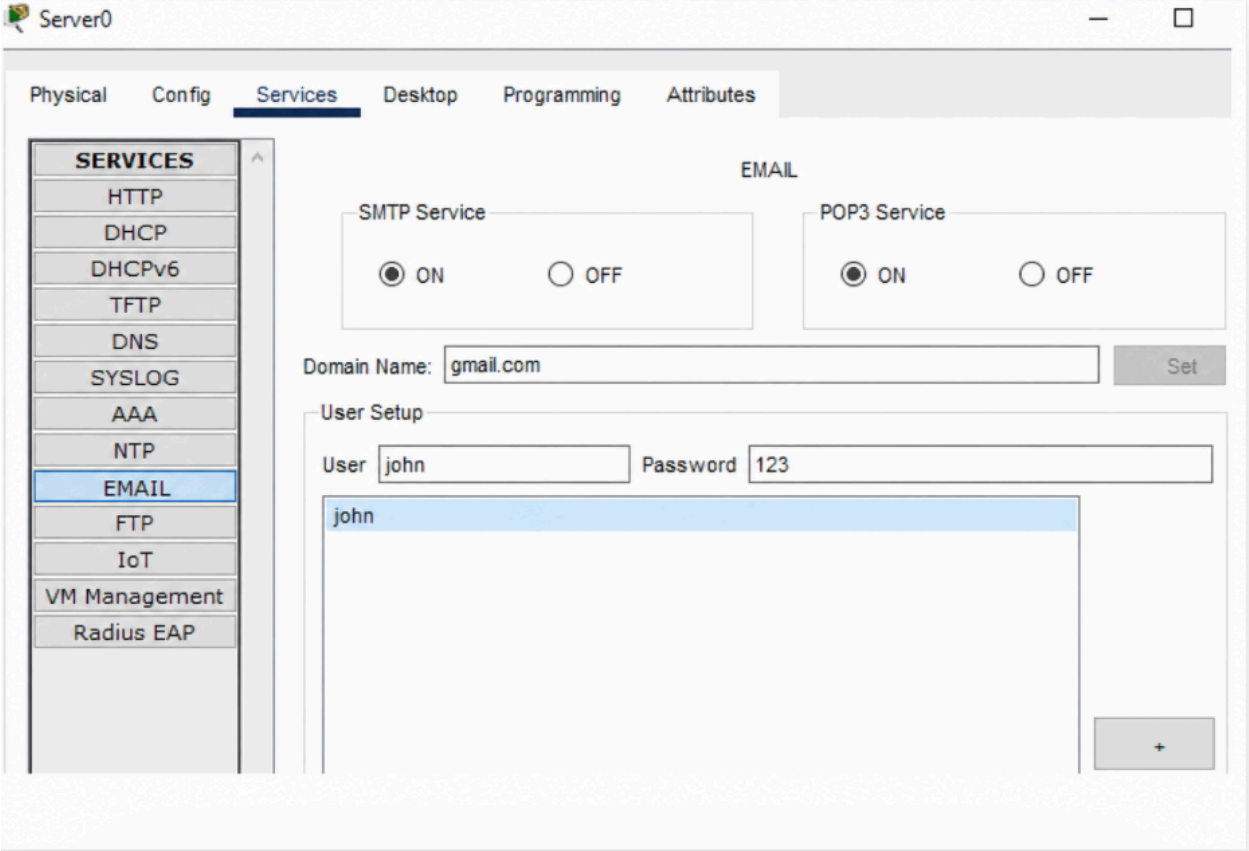
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

a(config)#username admin secret pass
*Mar 1 0:49:14.775: %SSH-5-ENABLED: SSH 1.99 has been enabled
a(config)#line vty 0 4
a(config-line)#login local
a(config-line)#trans in ssh
a(config-line)#exit
a(config)#ip ssh ver 2
a(config)#do wr
Building configuration...
[OK]
a(config)#
```

CopyPaste



4.



Server0

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name

Type

A Record

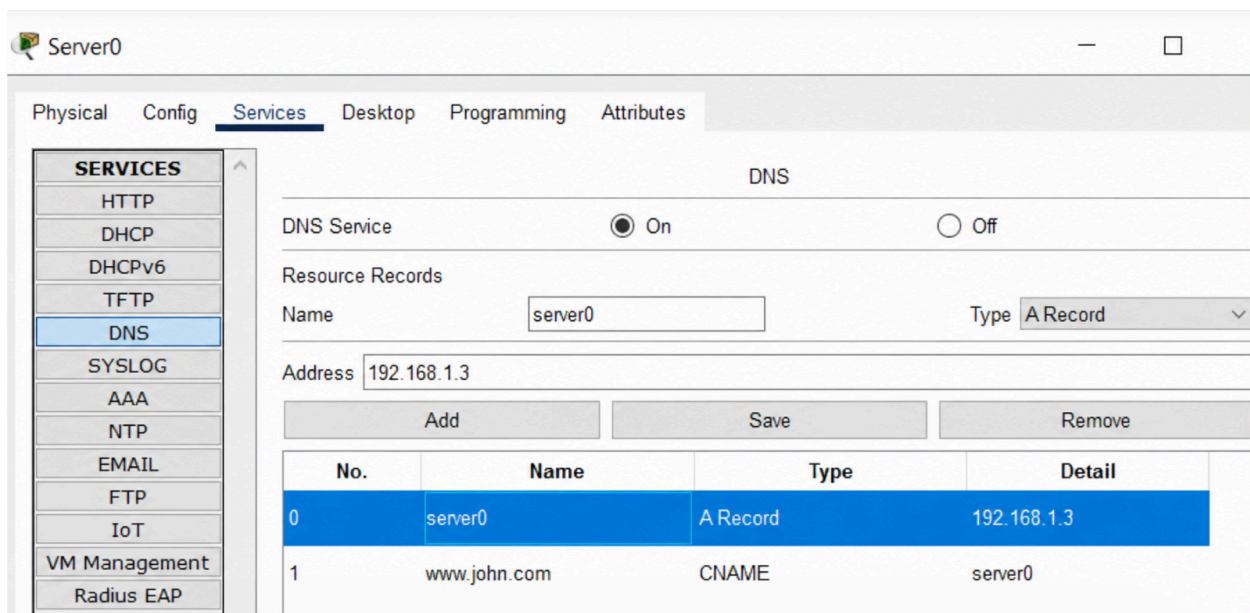
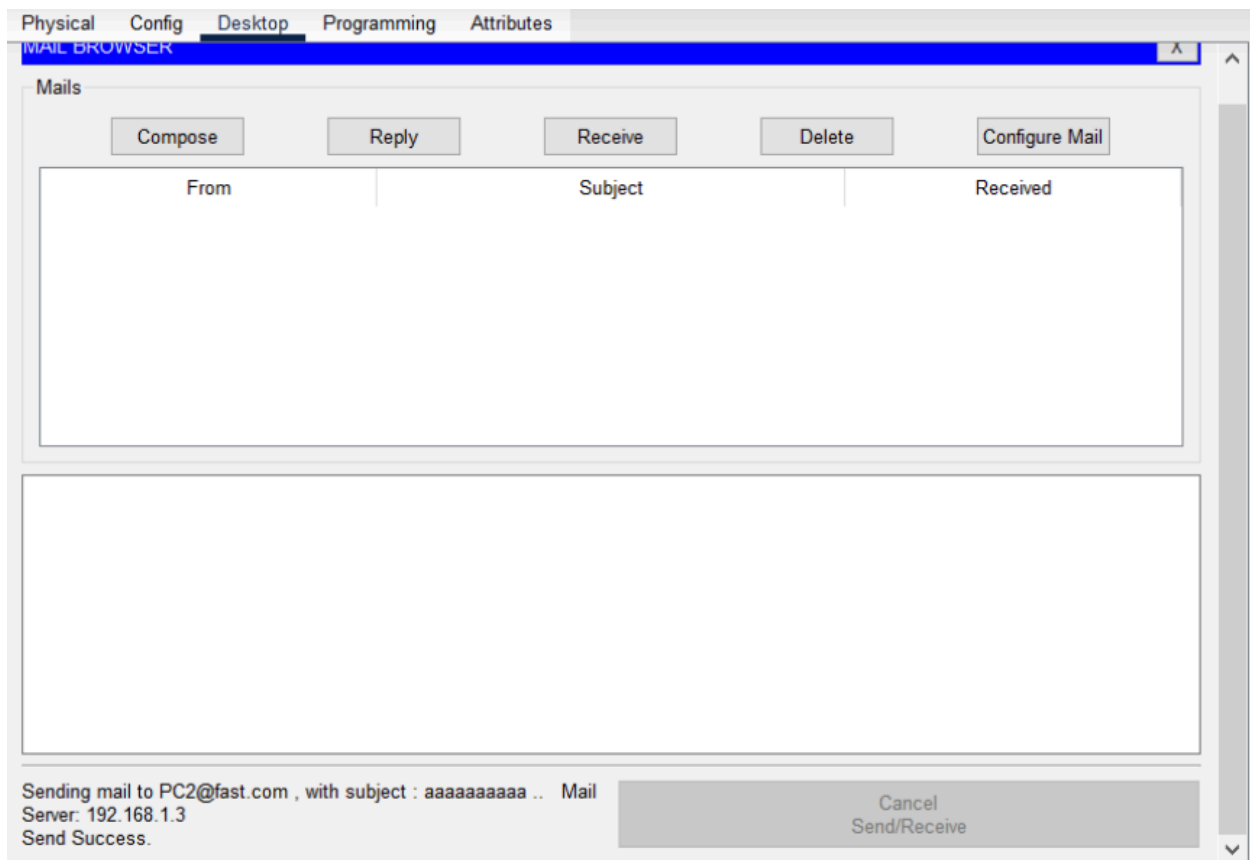
Address

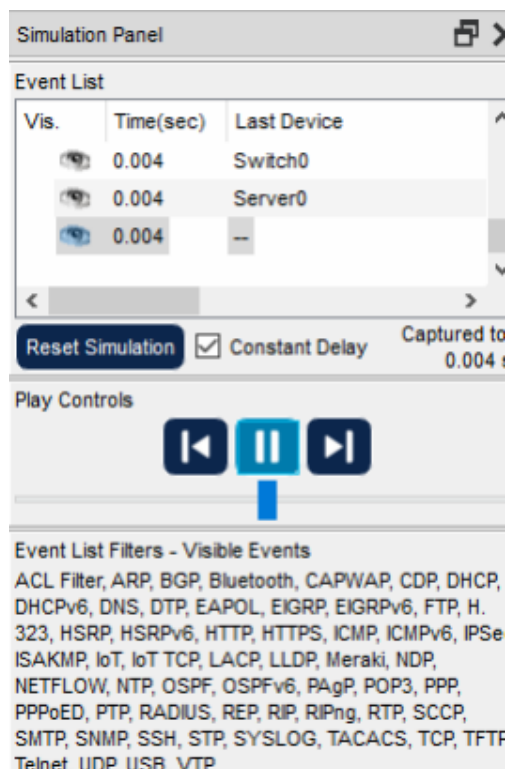
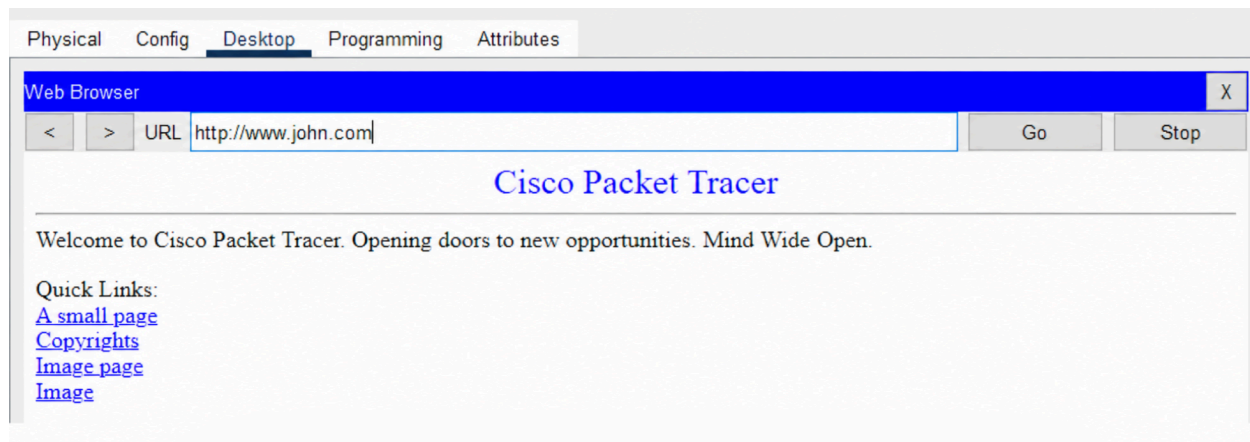
Add

Save

Remove

No.	Name	Type	Detail
0	gmail.com	A Record	192.168.1.20
1	nu.edu.pk	A Record	192.168.1.20





lab ex 2

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname router0
router0(config)#int fa0/0
router0(config-if)#ip address 192.168.1.1 255.255.255.0
router0(config-if)#ip nat inside
router0(config-if)#no shutdown

router0(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

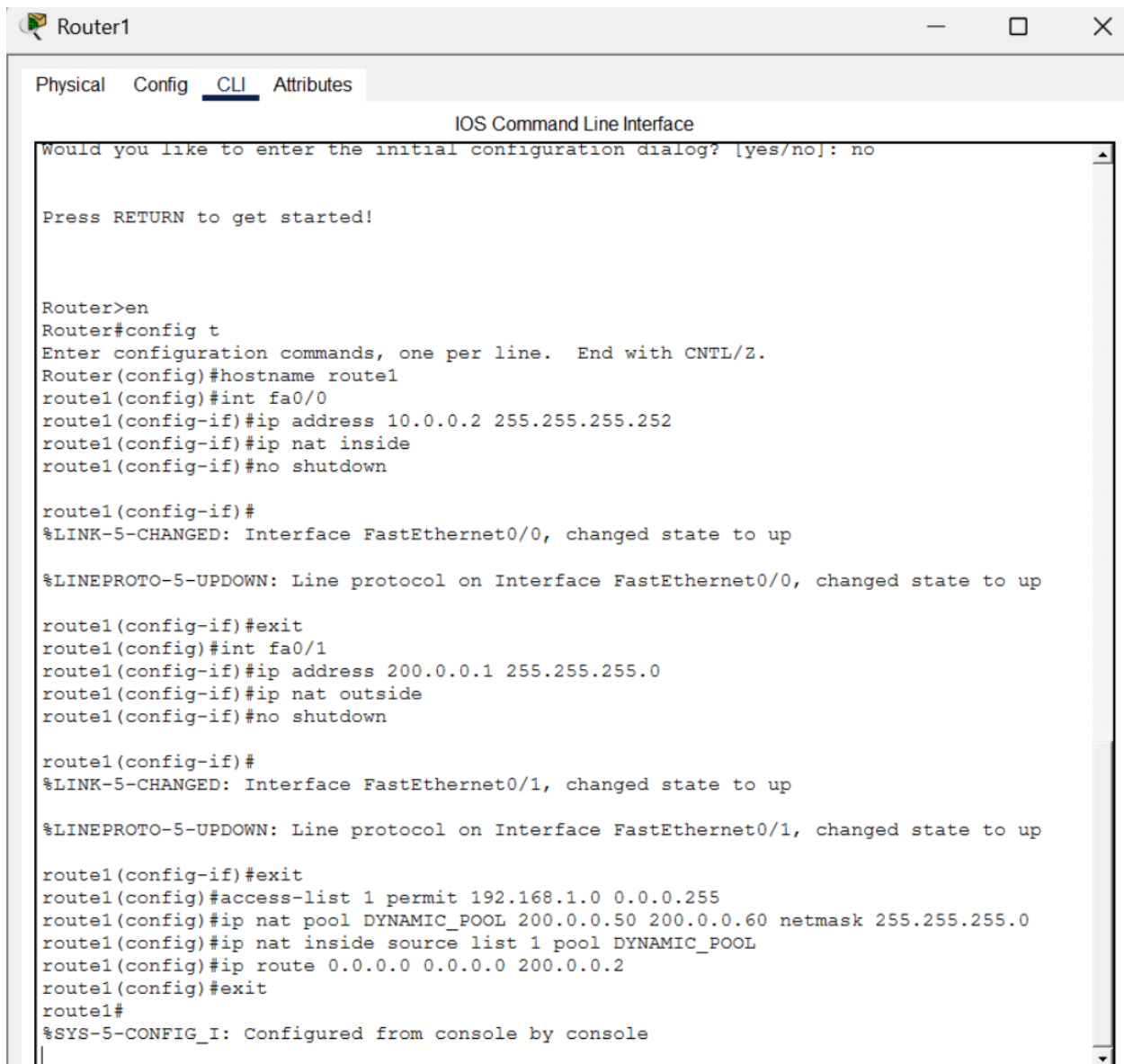
router0(config-if)#exit
router0(config)#int fa0/1
router0(config-if)#ip address 10.0.0.1 255.255.255.252
router0(config-if)#ip nat outside
router0(config-if)#no shutdown

router0(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

router0(config-if)#exit
router0(config)#ip nat inside source static 192.168.1.100 200.0.0.100
router0(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
router0(config)#exit
router0#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname route1
route1(config)#int fa0/0
route1(config-if)#ip address 10.0.0.2 255.255.255.252
route1(config-if)#ip nat inside
route1(config-if)#no shutdown

route1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

route1(config-if)#exit
route1(config)#int fa0/1
route1(config-if)#ip address 200.0.0.1 255.255.255.0
route1(config-if)#ip nat outside
route1(config-if)#no shutdown

route1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

route1(config-if)#exit
route1(config)#access-list 1 permit 192.168.1.0 0.0.0.255
route1(config)#ip nat pool DYNAMIC_POOL 200.0.0.50 200.0.0.60 netmask 255.255.255.0
route1(config)#ip nat inside source list 1 pool DYNAMIC_POOL
route1(config)#ip route 0.0.0.0 0.0.0.0 200.0.0.2
route1(config)#exit
route1#
%SYS-5-CONFIG_I: Configured from console by console
```

Server1

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

Server0

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

Router1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Static Routes

Network

Mask

Next Hop

Add

Network Address

192.168.2.0/24 via 192.168.2.2

0.0.0.0/0 via 192.168.2.2

Remove

Equivalent IOS Commands

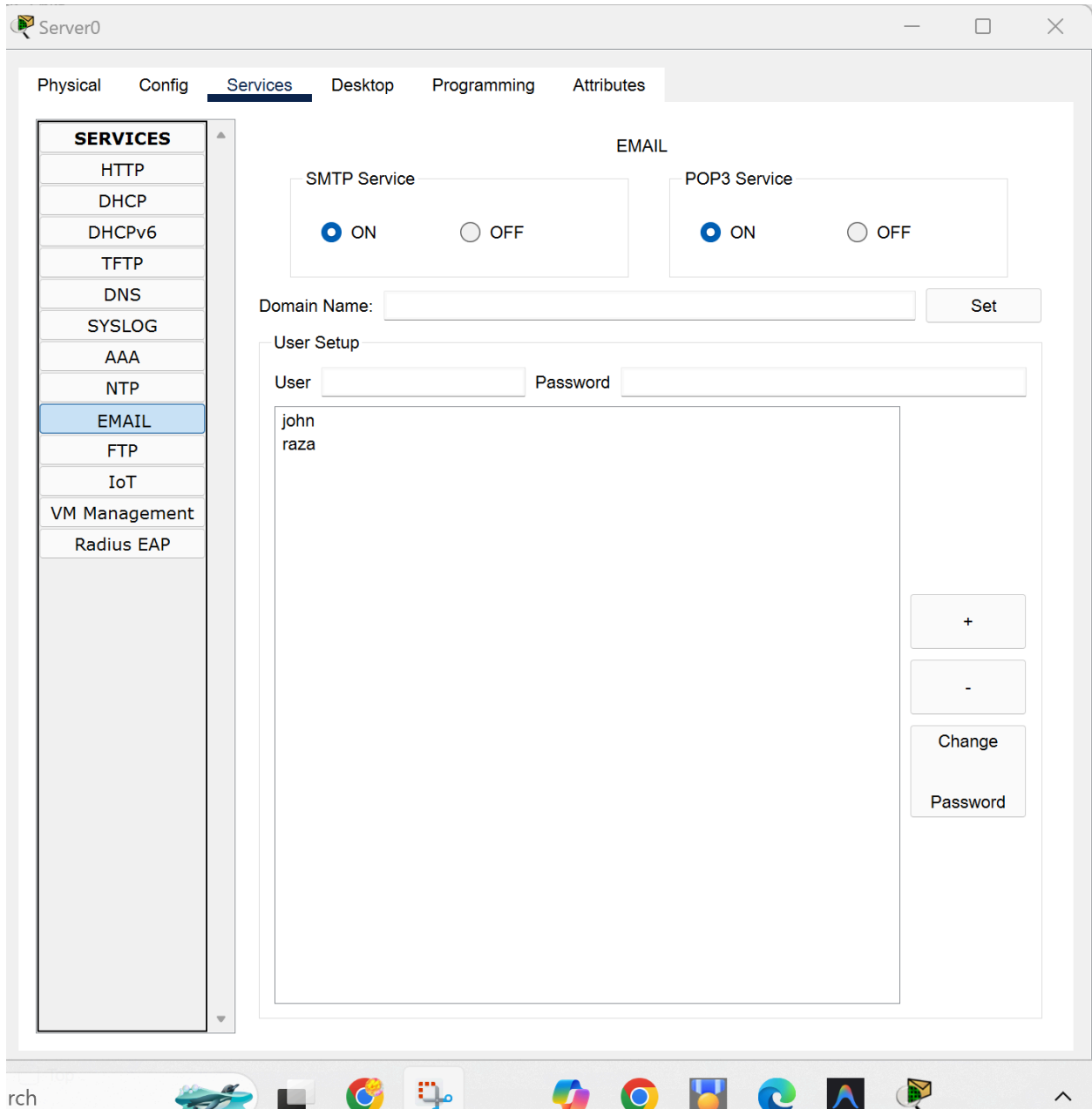
lab 9 ta... Router2

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface fa0/0
Router(config-if)# ip nat inside
Router(config-if)#
Router(config-if)# ip nat inside
Router(config-if)# ip nat inside
Router(config-if)#interface se2/0
Router(config-if)#ip nat outside
Router(config-if)#ip nat pool rightpool 192.168.2.20 192.168.2.30 netmask 255.255.255.0
Router(config)#access-list 1 permit 192.168.3.0 0.0.0.255
Router(config)#ip nat inside source list 1 pool rightpool
Router(config)#ip nat inside source list 1 pool rightpool
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.2.1
Router(config)#
```

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Physical Config **Services** Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

EMAIL

SMTP Service

☒ ON ☐ OFF

POP3 Service

☒ ON ☐ OFF

Domain Name: task2lab.com

Set

User Setup

User pc2 Password 456

pc1

pc2

+

-

Change

Password

☐ Top

SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **A Record**

Address

Add Save Remove

No.	Name	Type	Detail
0	mail.task2lab.com	A Record	192.168.1.100

DNS Cache

PC0

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: john

Email Address john@raza.com

Server Information

Incoming Mail Server 192.168.1.2

Outgoing Mail Server 192.168.1.2

Logon Information

User Name: john

Password: ●●●

Save Remove Clear Reset

PC2

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: raza

Email Address: raza@john.com

Server Information

Incoming Mail Server: 192.168.1.2

Outgoing Mail Server: 192.168.1.2

Logon Information

User Name: raza

Password: ●●●

Save Remove Clear Reset

Server1

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service

On

Off

Resource Records

Name

Type

A Record

Address

Add

Save

Remove

No.	Name	Type	Detail
0	raza.com	A Record	192.168.1.2

DNS Cache